

CHENGTAI RICHARD LI

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EDUCATION

University of California, Los Angeles (UCLA)

B.S. in Statistics and Data Science; Minors in Mathematics and Data Science Engineering

Los Angeles, CA

Expected June 2028

- GPA: 3.94/4.00 | Awards: UCLA Statistics Datathon 2025 – Award for Excellence in Analysis; Dean's Honor List
- Relevant Coursework: Data Structures and Algorithms, Machine Learning, Statistical Inference, Regression Analysis, Experimental Design, Optimization

EXPERIENCE

Data Engineer Intern

May 2025 – Aug 2025

Toronto, Canada

- Trained an Alternating Least Squares matrix-factorization model in PySpark to rank 50,000+ internal datasets for 500+ analysts on [REDACTED], an internal Tableau/Trino/Superset datahub.
- Productionized model outputs through a daily ETL pipeline delivering recommendations into the [REDACTED] UI, reducing manual dataset search time by 80%.
- Built daily Airflow and PySpark pipelines ingesting 3M+ rows across 5 internal sources, with data-quality validation so analysts had verified data each morning.

Full Stack Software Engineer

Mar 2026 – Present

Los Angeles, CA

[REDACTED] — 1 of 4 founding engineers, 10-person product team

- Own the Google Calendar integration end-to-end in a React Native (Expo, TypeScript) and Firebase app, syncing shared household events to personal calendars for 80+ households.
- Designed the Firestore schema and TanStack Query real-time sync layer, cutting document reads 45% and eliminating manual refresh across chores, calendars, and shared lists.
- Shipped a barcode pantry scanner integrating Open Food Facts (4M+ UPCs), Google Cloud Vision, and Firebase Cloud Functions, cutting grocery logging from 5 minutes of manual entry to under 30 seconds per haul.

PROJECTS

Scrubs | *Swift, SwiftUI, Core ML, Apple Vision, Zetic*

Apr 2026

- Won the Zetic Company Challenge at LA Hacks 2026 (UCLA) out of 1000+ competitors.
- Built an iOS app that detects and redacts PHI – names, DOBs, medical record numbers, faces – from healthcare photos entirely on-device, at 80ms per-image latency.
- Engineered an on-device vision pipeline combining Apple Vision OCR, a Zetic face detector, and a rule-based PHI classifier, validated across 1,000+ test photos with zero network calls.

Embers | *YOLOv11, Google Gemini, Next.js, Supabase, Flask, Python*

Apr 2025

- Won Best Use of Google Gemini and 1st Place, Cold Hard Cache Track at LA Hacks 2025 (UCLA) out of 500+ competitors.
- Built a video-to-inventory pipeline using YOLOv11 on a 2-minute home walkthrough, applying multi-frame deduplication and non-max suppression to prevent double-counting across overlapping frames.
- Integrated Google Gemini 2.5 for multimodal valuation, passing cropped item snapshots through a reasoning prompt inferring brand, condition, and market value for per-item replacement estimates.

LEADERSHIP

Athletics Director

2025 – Present

Los Angeles, CA

- Built the board's internal reimbursement and stipend portal (React, Supabase, Cloudflare Workers), used by 24 board members to submit, approve, and track reimbursements.
- Run athletics operations for a 1000+ member cultural student organization, delivering 4 events per quarter against a \$10K annual budget.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, Java, C++, R, Swift, Bash

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Asynchronous & Real-Time Systems

Data & Distributed Systems: PySpark, Apache Airflow, ETL/ELT, Distributed Data Processing, Data Modeling, Tableau

ML & Statistics: PyTorch, scikit-learn, NumPy, Pandas, OpenCV, Collaborative Filtering, Regression Analysis, A/B Testing, Experimental Design

Frameworks & APIs: React, React Native, Next.js, Node.js, SwiftUI, Flask, Tailwind CSS, REST API Design & Integration

Cloud & Infrastructure: AWS, Firebase, Supabase, Cloudflare Workers, Serverless/Cloud Functions, Docker, Git, CI/CD, PostgreSQL, Trino